

SILDENAFIL CITRATE (VIAGRA) INDUCED ENHANCEMENT OF RIGHT ATRIAL CONTRACTILITY IN ISOLATED RAT HEART IN VITRO

NABEEH I. A. RAKKAH*

Department of Physiology, Faculty of Medicine and Medical Sciences,
Umm-al-Qura University, Makkah, Saudi Arabia

ABSTRACT

The present study deals with the chronotropic effects of Sildenafil citrate (Viagra) on right atrial contractility of mammalian isolated heart muscle.

Effects of Sildenafil citrate on rat isolated heart *in-vitro* using right atrial samples, in comparison with nor-adrenaline, revealed that Sildenafil at 10^{-7} M and above produced dose related increases in contractility. The maximum observed increase at 10^{-4} M represented a $126\% \pm 31\%$ increase above control amplitude. Nor-adrenaline produced similar dose-related increases in contractility and was approximately three times more potent than Sildenafil citrate. Dimaprit also produced similar responses to the Sildenafil but was less potent. These results indicated that rat heart responds to Sildenafil citrate, which increases contractility and enhances atrial automaticity, with sensitivity comparable to that for nor-adrenaline. The selective antagonism of the Sildenafil contractility response by cimetidine indicated that this response is mediated via histamine H₂-receptors and not H₁ noradrenergic β receptors. The involvement of histamine receptors is confirmed by the similarity of the response to dimaprit. The production of spontaneous contractions by dimaprit further suggests that Sildenafil citrate-induced enhancement of atrial automaticity is mediated via modified histamine H₂-receptors.

INTRODUCTION

Sildenafil belongs to a class of compounds called phosphodiesterases (PDEs) inhibitors. PDEs comprise a diverse family of enzymes that hydrolyze cyclic adenosine monophosphate (cAMP) and cyclic guanosine monophosphate (cGMP) and therefore play a critical role in the modulation of second-messenger signaling pathways (Beavo, 1995). By inhibiting the hydrolytic breakdown of cGMP, Sildenafil prolongs the action of cGMP. Sildenafil potentiates the activity of cGMP in the corpus cavernosum, thereby augmenting vasodilator activity of neuronally mediated nitric oxide production. Sildenafil has also been demonstrated to increase cGMP levels and cause smooth muscle relaxation in isolated segments of epicardiac coronary artery. However, the effect of Sildenafil on

coronary resistance vessels has not been studied (Rajfer *et al.*, 1992).

Sildenafil is a potent and selective inhibitor of cGMP-specific PDE5, PDE3 and PDE2, the predominant isozymes that metabolizes cGMP in the corpus cavernosum of the penis, in vascular, visceral, tracheal smooth muscles, cardiac muscles and in platelets (Naylor 1998).

Over 4000-fold greater selectivity for PDE5 over PDE3 is important because inhibitors of PDE3 (the isozyme involved in regulation of cardiac contractility), such as milrinone, vesnarinone, and enoximone, that have been used in patients with heart failure are generally associated with increased incidence of cardiac arrhythmias and other serious side effects (Packer *et al.*, 1991).

Unlike cAMP-specific PDE3 inhibitors (milrinone, vesnarinone, and enoximone) that

*Correspondence

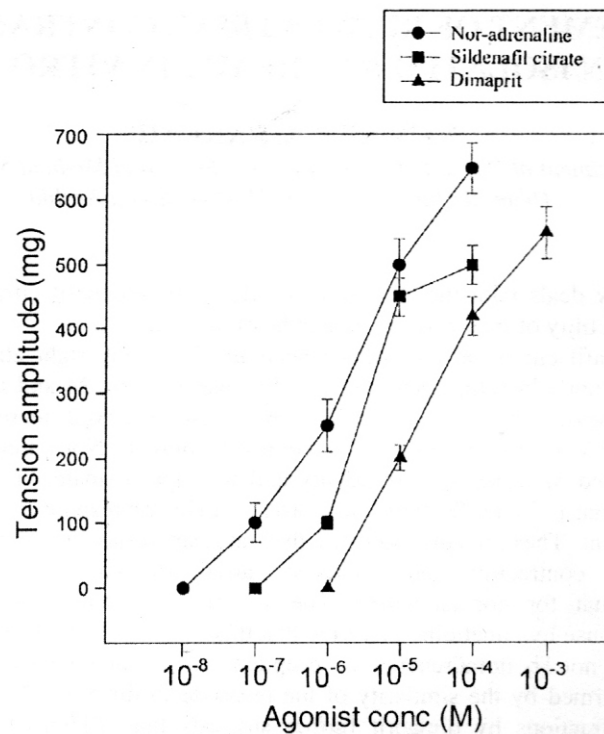


Fig. 1: Effect of Sildenafil citrate, nor-adrenaline and dimaprit on rat right atrial contractility (n=10). Points are mean $\pm$ standard errors. n = indicates sample number.

increase long-term mortality in patients with heart failure (Nony *et al.*, 1994), Sildenafil is highly selective (>4000-fold), for human PDE5 over human PDE3 and has not been found to elevate cAMP (Egan *et al.*, 2002). The cardio-toxic effects of PDE3 inhibitors are thought to be related to increases in intracellular cAMP in the myocardium (Degerman *et al.*, 1997). Further more, PDE5 is not present in cardiac myocytes, and Sildenafil has been shown to have no direct inotropic effects on dog trabeculae muscle (Kyung *et al.*, 2002). However, Sildenafil citrate has not been investigated extensively in heart failure patients.

Sildenafil produces a transient modest reduction in systolic (8-10 mmHg) and diastolic (5-6 mmHg) blood pressure, with peak effects evident at 1 hour after the dose (coincident with peak plasma concentrations)

and returning to baseline values by 4 hours after the dose (Jiang *et al.*, 2002).

The hypotensive effects of Sildenafil are neither age dependent nor close related and may rarely result in reports of orthostatic effects. In normal human volunteers, no significant changes in cardiac index are evident up to 12 hours after the dose for oral Sildenafil (100-200mg), or intravenous Sildenafil (20-80mg) (Hoepfer *et al.*, 2002). Significant decreases in systemic vascular resistance index are reported at the end of intravenous Sildenafil infusion (20-80mg), when plasma concentrations were highest. Sildenafil has both arteriodilator and venodilator effects on the peripheral vasculature. In patients with angina, intravenous Sildenafil has been found to reduce systemic and pulmonary arterial pressure and cardiac output with its mixed

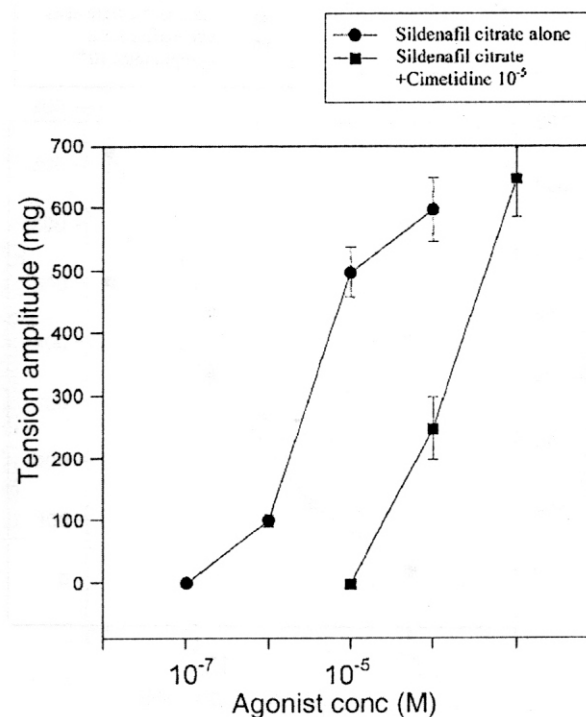


Fig. 2A: Effect of Sildenafil citrate on rat right atrial contractility alone and in the presence of cimetidine 10^{-5} M. (n=10). Points are mean $\pm$ standard errors.
n = indicates sample number.

arterial and venous vasodilator effects, however controversial effects are observed on heart rate in different animal species and on humans (Arora *et al.*, 1999).

In conclusion, consistent with the anticipated effects resulting from an increase in cGMP levels in vascular smooth muscle, Sildenafil possesses vasodilatory properties, which affects cardiac contractility, blood pressure and heart rate, central hemodynamic and peripheral vasculature, although the effects are mild and generally clinically insignificant, when taken alone.

In a drug interaction study of Sildenafil and amlodipine, the additional blood pressure reduction in the patient population receiving both Sildenafil and amlodipine have been found not significantly different from the

population receiving Sildenafil and a placebo. Although drug-drug interaction studies have not been conducted with the following medications, no increase in blood pressure-related adverse events or systemic enhancement of blood pressure-lowering effects of thiazide, loop and potassium-sparing diuretics, acetylcholine inhibitors, calcium channel blockers, or alpha or beta adrenergic receptor antagonists have been observed in clinical trials (Stief *et al.*, 2000) and there may be a theoretical concern in the patients receiving multiple medications that include anti hypertensive therapy and an inhibitor of the metabolic pathway (cytochrome P450 3A4) of Sildenafil citrate, and more research needs to be done to assess the specific risks of Sildenafil citrate (Viagra) use among cardiovascular studies.

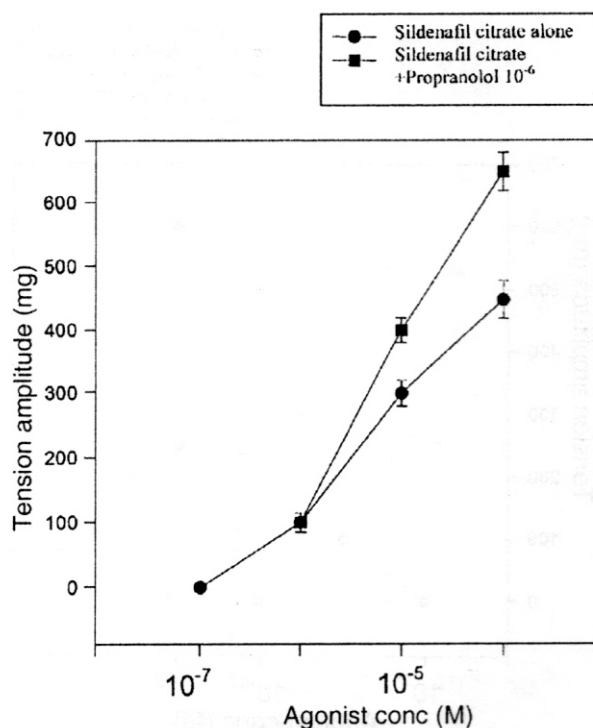


Fig. 2B: Effect of Sildenafil citrate on rat right atrial contractility alone and in the presence of cimetidine 10^{-6} M. (n=10). Points are mean $\pm$ standard errors.
n = indicates sample number.

MATERIALS AND METHODS

Chronotropic effects of the Sildenafil citrate on the rat isolated heart muscle in vitro were investigated using ten right atrial samples using adult, sexually active male Sprague-Dawley (450-500gm.) rats by established methods (Drolet *et al.*, 1998). Immediately after excision, the samples were placed in modified Krebs-Henseleit buffer solution prepared with distilled deionized (Millipore) water (Peter *et al.*, 2000) at 4°C , then transferred to solution equilibrated with 5% CO_2 in O_2 at ambient temperature. Right atrial trabeculae carneae 0.94 ± 0.03 cm X 0.103 ± 0.03 cm, were mounted vertically in a 60 ml Harvard single tissue glass organ bath containing modified Krebs-Henseleit buffer solution equilibrated with 5% CO_2 in O_2 at 37.5°C . The lower end of the preparation was

connected to a hook provided with a stainless steel rod, while the other end was connected to the research grade Harvard isometric displacement transducer model AH 60-2997 by means of strong cotton thread. This tissue which normally did not exhibit spontaneous contraction was placed under 1 g. resting tension and isometrically stimulated to contract with a pulse of supramaximal voltage (usually 20-30 V) and of 2mS duration using a Grass stimulator (Model S88, USA). Isometric tensions were recorded on a Harvard 2-channel modular Universal Oscillograph through amplifier interface model All 50-8861. The preparations were allowed sufficient time to develop a stable contraction (approximately 45 minutes) during which the bath fluid was continuously over flowed. The effects of the Sildenafil citrate on contractility (tension amplitude)

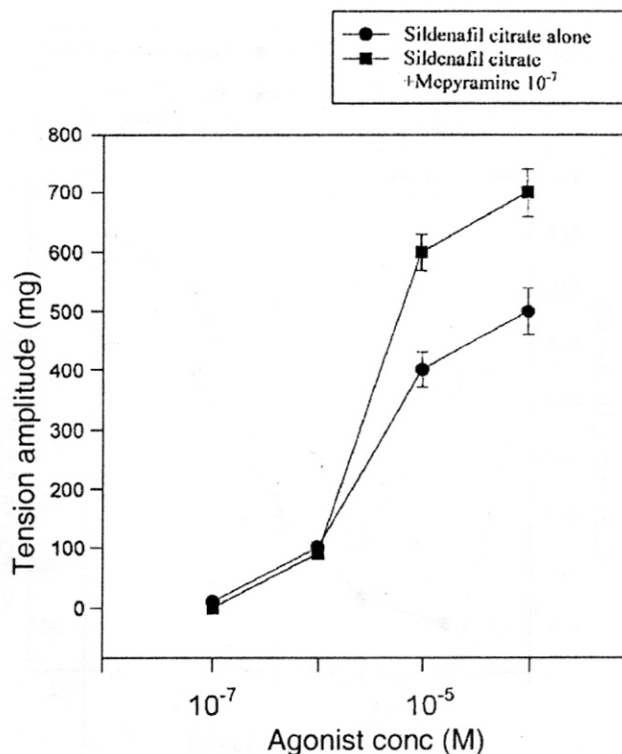


Fig. 2C: Effect of Sildenafil citrate on rat right atrial contractility alone and in the presence of cimetidine 10^{-7} M. (n=10). Points are mean $\pm$ standard errors.
n = indicates sample number.

were compared with nor-adrenaline and the selective histamine H_2 -receptor agonist dimaprit by obtaining cumulative dose response curves to each agonist. Three dose-response curves were obtained in each preparation of this study with the agonist administration order assigned at random. Drug solutions were added to the bath in a volume not exceeding 0.1 ml and responses allowed to reach a stable maximum before increasing the bath concentration. After completion of dose response curves tissues were washed by overflow and allowed 15-30 minutes to recover. The effects of histamine receptor antagonist's cimetidine (H_2), mepyramine (H_1) or the adrenergic p-receptor antagonist propranolol on the Sildenafil citrate and nor-adrenaline responses were also investigated. For this study control dose

response curves to Sildenafil and nor-adrenaline in random order were first obtained in each preparation. The antagonist was then added to the perfusion medium and an equilibration period of at least 15 minutes allowed during which the bath fluid was changed at least every 5 minutes. Sildenafil and nor-adrenaline dose response curves were then repeated in the same order as that before antagonist administration. Four dose-response curves were thus obtained in each preparation for the antagonist study.

Drugs used were Sildenafil citrate (Pfizer), cimetidine (Sigma), dimaprit-dihydro chloride (Sigma), mepyramine maleate (Sigma), nor-adrenaline bitartrate (Sigma,) and propranolol hydrochloride (Sigma).

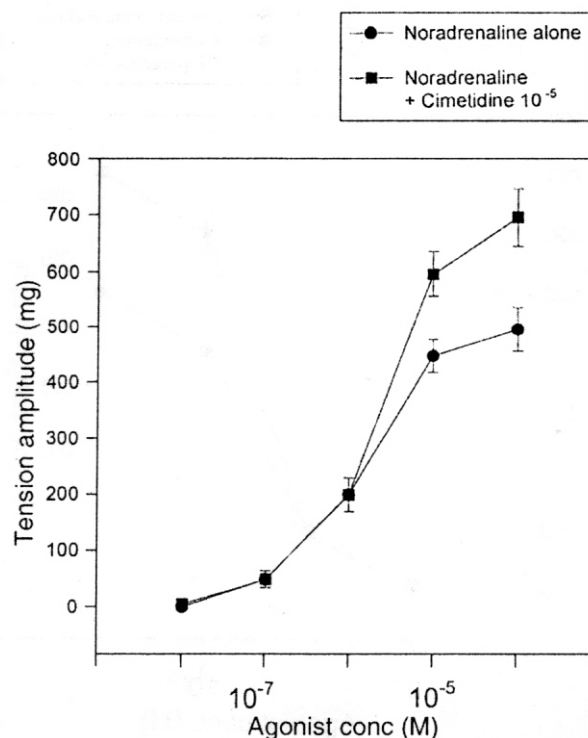


Fig. 3A: Effect of Sildenafil citrate on rat right atrial contractility alone and in the presence of cimetidine 10^{-5} M. (n=10). Points are mean $\pm$ standard errors.

n = indicates sample number.

RESULTS

Effects of Sildenafil citrate (Viagra), nor-adrenaline and dimaprit on rat right atrial contractility are summarized in Figure 1. These results revealed that the Sildenafil at 10^{-7} M and above produced dose related increases in right atrial contractility. The maximum observed increases at 10^{-4} M represented a $126\% \pm 31\%$ increase above control amplitude (n=10). Nor-adrenaline produced similar dose-related increases in contractility and was approximately 3 times more potent than the Sildenafil citrate. Dimaprit also produced similar responses to Sildenafil citrate and was less potent. All these agonist produced similar maxima. The Sildenafil dose-response curve repeated in the presence of Cimetidine 10^{-5} M lay to the right of control indicating antagonism by

Cimetidine of the Sildenafil response whereas curves repeated in the presence of either mepyramine 10^{-7} M or propranolol 10^{-6} M were similar to control (Figures 2A, B, C). In contrast, the nor-adrenaline dose response curve repeated in the presence of Cimetidine was similar to control, whereas the curve repeated in the presence of propranolol lay to the right of control, indicating antagonism by propranolol of the nor adrenaline response (Figures 3A and B). The nor-adrenaline dose-response curve repeated in the presence of mepyramine like extract was similar to the control (Figure 3C).

DISCUSSION

Sildenafil citrate produces large changes in cardiac function in many animal species both *in viva* and *in-vitro*. The cardiac

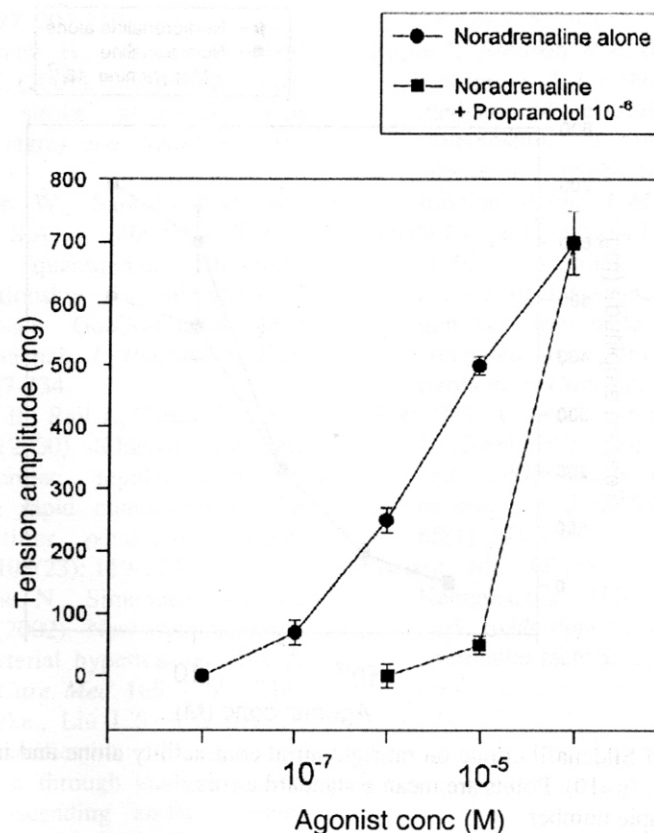


Fig. 3B: Effect of Sildenafil citrate on rat right atrial contractility alone and in the presence of cimetidine 10^{-6} M. (n=10). Points are mean $\pm$ standard errors.
n = indicates sample number.

response to Sildenafil in the guinea-pig *in vitro* induces increases in sinus rate, force of ventricular contraction, coronary flow, aortic flow, total cardiac output, and external pressure-volume work (Geelen *et al.*, 2000). The above cardiac actions of Sildenafil are all competitively antagonized by histamine H₃-receptor antagonists and therefore involve stimulation of histamine H₃-receptors (Salata *et al.*, 1995; Ekins *et al.*, 2002), little information is however documented regarding the stimulation of histamine H₂ receptors (Swissa *et al.*, 2002). In the present study we have therefore investigated the histaminergic H₂ stimulatory effects of Sildenafil citrate (Viagra), nor-adrenaline and

dimaprit on rat right atrial muscles *in vitro*.

Our results showed spontaneous contractions produced by Sildenafil citrate, nor-adrenaline and dimaprit. The spontaneous rate was not quantified; however, the incidence of spontaneous activity, and the drug concentration at which it first occurred was noted. Sildenafil up to 10^{-4} M produced spontaneous activity in 7/24 (29%) preparations. Activity was first noted at 10^{-6} M in one preparation (4%) and by 10^{-5} M spontaneous activity occurred in 5/24 (21%) preparations. Nor-adrenaline up to 10^{-4} M produced spontaneous activity in 9/19 (47%) preparations. Activity was first noted at 10^{-6} M

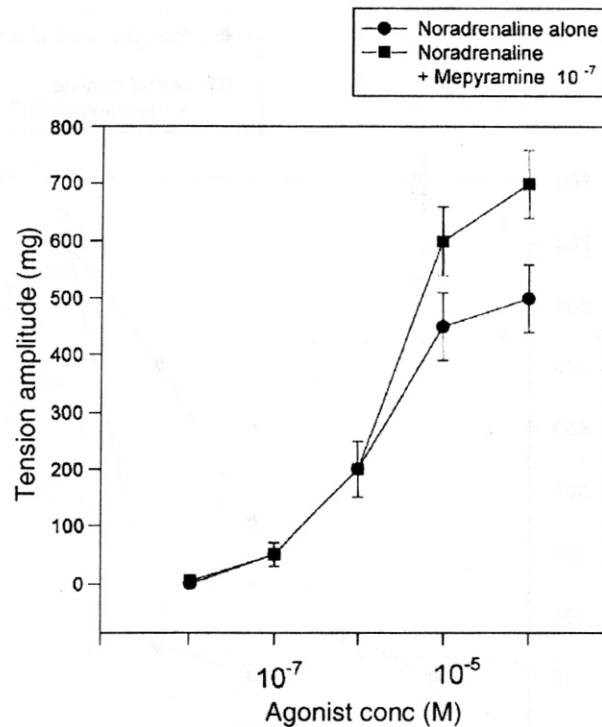


Fig. 3C: Effect of Sildenafil citrate on rat right atrial contractility alone and in the presence of cimetidine 10^{-7} M. (n=10). Points are mean $\pm$ standard errors.
n = indicates sample number.

in one (5%) preparation and by 10^{-5} M activities had occurred in 6/19 (32%) preparations. Dimaprit up to 10^{-4} M produced spontaneous activity in 1/4 (25%) preparations. These results indicate that the rat heart responded to Sildenafil citrate that increases contractility and enhances atrial automaticity with sensitivity comparable to that for nor-adrenaline. The selective antagonism of the histaminergic contractility response by Cimetidine indicates that this response is mediated via histamine H_2 -receptors and not H_1 nor-Adrenergic β -receptors. The involvement of H_2 -histamine receptor was confirmed by the similarity of the response to dimaprit. The production of spontaneous contractions by dimaprit further suggests that Sildenafil citrate (Viagra) induced enhancement of atrial automaticity is mediated via histamine H_2 -receptors.

REFERENCES

- Arora R.R., Timoney M., Melilli L. (1999). Acute myocardial infarction after the use of Sildenafil. *N. Engl. J. Med.* **341**: 700-710.
- Beavo J.A. (1995). Cyclic nucleotide phosphodiesterases: functional implications of multiple isoforms. *Physiol. Rev.* **75**: 725-748.
- Degerman E., Belfrage P. and Manganiello V.C. (1997). Structure, localization, and regulation of cGMP-inhibited phosphodiesterase (PDE3). *J. Biol Chem.* **272**: 6823-6826.
- Drolet B., Khalifa M., and Daleau P. (1998). Block of rapid component of the delayed rectifier potassium current by the prokinetic agent cisapride underlies drug-

- related lengthening of the QT interval. *Circulation*. **97**: 204-210.
- Egan R., Ponieranz H., Mogan J.C. and Johnston K.C. (2002). Transient ischemic attack and stroke associated with Sildenafil (Viagra) use. *Neurology*. **59**: 293-294.
- Ekins S., Crumb W., Sarazen R.D. and Wrighton S.A. (2002). Three-dimensional quantitative structure-activity relationship for inhibition of human Eter-a-a Go-Go-related gene potassium channel. *J. Pharmacol Exp. Ther.* 301: 427-434.
- Geelen P., Drolet B., Rail J., Cardinal R. and Turgeon J. (2000). Sildenafil (Viagra) prolongs cardiac repolarization by blocking the rapid component of the delayed rectifier potassium current. *Circulation*. **103(23)**: 119-120.
- Hoepfer M., Galie N., Simonneau G. and Rubin L.J.. (2002). New treatments for pulmonary arterial hypertension. *Am. J. Respir. Crit. Care. Med.* **165**: 1209-1216.
- Jiang X., O'Rourke., Liu L.S. and Liu SZ (2002). Quantification of glyceryl trinitrate effect through analysis of the synthesized ascending aortic pressure waveform. *Heart*. **88**: 143-148.
- Kyung Y., Has K., Jai D. and Jong L. (2002). Sildenafil (Viagra) augments sodium nitroprusside-induced hypotension in dogs. *Anesth. Analg.* **94(6)**: 1505-1513.
- Naylor A.M. (1998). Endogenous neurotransmitters mediating penile erection. *Br. J. Urol.* **81**: 424-43.
- Nony P., Boissel J.P. and Lievre M. (1994). Evaluation of the effect of phosphodiesterase inhibitors on mortality in chronic heart failure patients: a meta-analysis. *Eur. J. Clin. Pharmacol.* 46:191-196.
- Packer M., Carver J.R. and Rodehoffer R.J. (1991). Effects of oral milrinone on mortality in severe chronic heart failure. *N. Engl. J. Med.* **325**:1468-1475.
- Peter G., Benoit D., Jimmy K., Jocelyn B. and Jacques T. (2000). Sildenafil (Viagra) prolongs cardiac repolarization by blocking the rapid component of the delayed rectifier potassium current. *Circulation*. **102(3)**: 275-282.
- Rajfer J., Aronson W.J., Bush P.A., Dory F.J. and Lgnarro L.J. (1992). Nitric oxide as a mediator of relaxation of the corpus cavernosum in response to non-adrenergic, noncholinergic neurotransmission. *N. Eng. J. MED.* **326**: 90-94.
- Salata J.J., Jurkiewicz N.K. and Wallace A.A. (1995). Cardiac electrophysiological actions of histamine H₁-receptor antagonists astemizole and terfenadine compared with chlorpheniramine and pyrilamine. *Circ. Res.* **76**: 110-119.
- Stief C.G., Uckert S., Becker A.J. and Jonas U. (2000). Effects of Sildenafil on cAMP and cGMP levels in isolated human cavernous and cardiac tissues. *Urology*. **55(1)**: 146-150.
- Swissa M., Ohara T., Shah P.K. and Karagueuzian H.S. (2002). Sildenafil nitric oxide donor combination promotes ventricular tachyarrhythmias in the swine right ventricle. *Am. J. Physiol.* **282**: 1778-1791.